Yogesh Murala

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SUMMARY

Innovative Senior Engineer with 3 years of experience in applied machine learning and generative AI at Bosch, specializing in architecting LLM systems to reduce manual triage efforts by 60%. Developed scalable RAG pipelines and engineered AI-driven assistants to enhance retrieval accuracy and response reliability. Seeking to leverage skills in AI/ML to drive transformative production-ready AI solutions through strategic collaboration and cutting-edge technologies.

EXPERIENCE

Senior Engineer - Applied Machine Learning & Generative AI

Bosch Global Software Technologies — **August 2023 - Present**

- Architected and deployed production LLM systems to extract structured insights from unstructured engineering data(logs, JIRA tickets, release notes), reducing manual triage effort by 60
- Designed scalable RAG pipelines including intelligent chunking strategies, embedding model integration, vector databaseindexing, semantic search, and re-ranking for high-relevance retrieval.
- Built LLM-powered conversational assistants using structured prompt engineering and hybrid deterministic + generativereasoning to improve response reliability and reduce hallucinations.
- Implemented Databricks-based ingestion pipelines to centralize engineering data as canonical datasets, enabling periodicembedding refresh and consistent retrieval behavior.
- Developed and maintained Python REST APIs for LLM inference and embedding services with validation, logging,monitoring, and CI/CD integration.
- Orchestrated GenAI workflows using Apache Airflow, covering prompt experimentation, embedding lifecycle management,evaluation pipelines, and automated metric capture.
- Collaborated with product owners, architects, and stakeholders to translate business requirements into scalable, production-readyAI solutions.
- Mentored junior engineers on RAG architecture, prompt engineering best practices, and production ML system designwithin Scrum teams.
- Instituted robust monitoring and retraining workflows for production LLM systems, ensuring consistent model performance and rapid adaptation to evolving engineering data distributions.

Project Trainee - Machine Learning & Systems Optimization

Bosch Global Software Technologies — **September 2022 - July 2023**

- Optimized deep learning inference pipelines using Apache TVM (auto-tuning, operator fusion) and NVIDIA TensorRT,improving latency and throughput for real-time AI systems.
- Applied GPU-level optimizations including mixed-precision inference and CUDA acceleration to enhance performanceunder production constraints.
- Developed and deployed real-time computer vision models using PyTorch and OpenCV, supporting benchmarking andperformance analysis.
- Spearheaded a 6-month AI engineering initiative utilizing PyTorch and OpenCV to automate model benchmarking and deployment pipelines, directly reducing manual integration time by 40% for production-grade computer vision workflows.
- Engineered scalable ML platforms integrating PyTorch, Apache TVM, and NVIDIA TensorRT over a 6-month period to centralize model lifecycle management, supporting deployment and benchmarking of 8+ computer vision models daily while achieving a 30% reduction in end-to-end inference time.

EDUCATION

B.Tech in Computer Science Engineering (Artificial Intelligence)

Minor in CSE • KL University • Vijayawada, Andhra Pradesh • 2023 • 9.2/10

- Acted as a Peer Mentor where Mentored undergraduate students in Machine Learning, Deep Learning, Natural Language Processing, and Signal Processing
- Prepared laboratory manuals and guided peer-led sessions to help students understand practical implementations and assignments.

SKILLS

Programming: Python, C/C++, SQL

Generative AI & ML: LLMs, Transformers, RAG, Prompt Engineering, Fine-tuning (SFT, LORA), NLP

Frameworks: PyTorch, TensorFlow, LangChain, LangGraph, OpenCV

Cloud & Data: Microsoft Azure (Azure DevOps, Databricks), Vector Databases, Embeddings, ETL Pipelines, REST APIs

MLOps: Apache Airflow, MLflow, CI/CD Pipelines

Optimization: Apache TVM, TensorRT, CUDA